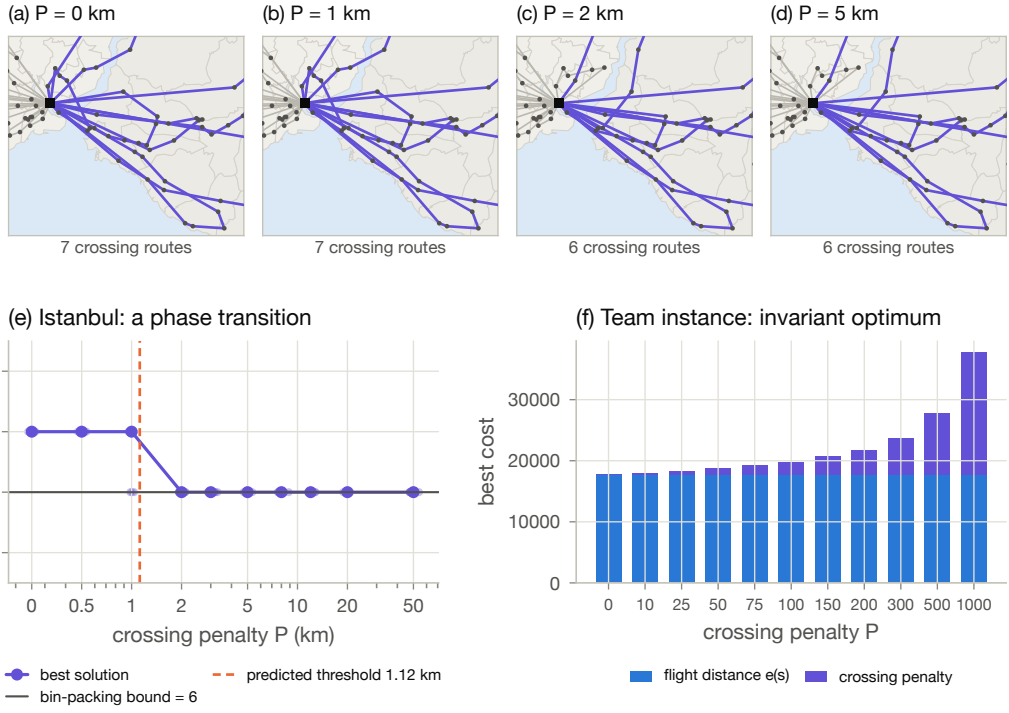

(a) P = 0 km
(b) P = 1 km
(c) P = 2 km
(d) P = 5 km
7 crossing routes
7 crossing routes
6 crossing routes
6 crossing routes
(e) Istanbul: a phase transition
routes crossing
8
7
6
5
0
0.5
1
2
5
10
20
50
crossing penalty P (km)
best solution
predicted threshold 1.12 km
bin-packing bound = 6
(f) Team instance: invariant optimum
best cost
30000
20000
10000
0
0
10
25
50
75
100
150
200
300
500
1000
crossing penalty P
flight distance e(s)
crossing penalty